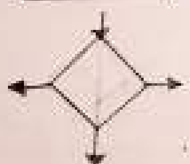
Basic Building Block -

- Things
  - Structural
  - Behavioural
  - Grouping
  - Annotational
- Relationship
  - Association
  - Dependency
  - Generalization
  - Realization
- Diagram
  - Structural
  - Behavioral

Initial node → ●

← Activity Name



← Decision node

Merge node

Experiment - 4

Aim:- To understand & model the flow of activities in a system, using UML activity Diagram and state machine diagram.

Theory -

Activity Diagram -

- An Activity Diagram represent the workflow of a system, showing the sequence of activities & decision-making paths.
- It is mainly used for business process modeling & detailed logic of use cases.
- It is similar to a flowchart but supports parallel & concurrent processes.

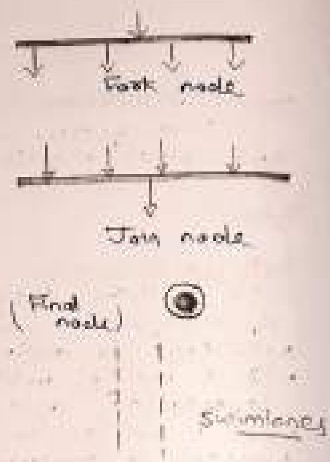
(a) Building Blocks of Activity diagram:-

(i) Initial node - The starting point of the activity (black filled circle)

(ii) Activity / Action - Represents a task or function (rounded rectangle)

(iii) Decision Node - Represent branching with conditions (diamond)

(iv) Merge Node - Combines multiple flows into one (diamond)



(v) **Fork node** - Splits one flow into multiple parallel flows (thick horizontal bar)
(Branching)

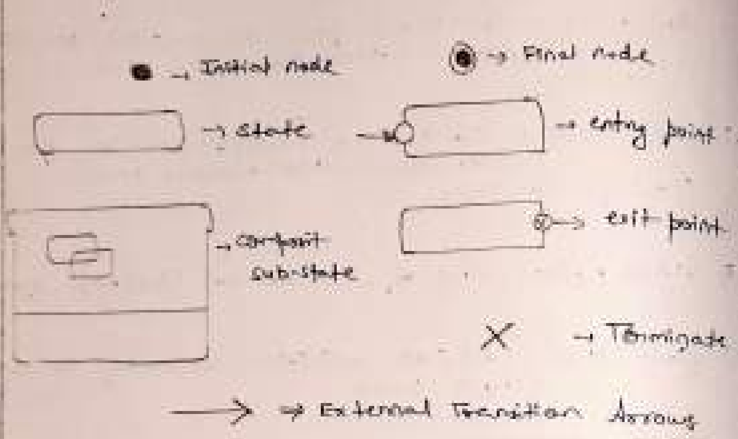
(vi) **Join node** - Synchronizes multiple parallel flows into one (thick horizontal bar)
(merge branching)

(vii) **Final node** - End of the process (bullseye symbol)

(ix) **Swimlanes** - Divide activities according to actors / roles.

~~State diagram:-~~

Procedure to draw an Activity Diagram:-
(1) Identify the starting point (Initial node)
(2) List all the activities or actions in sequences.
(3) Add decision points where branching occurs
(4) Use fork / Join for parallel tasks if required.
(5) Connect activities using control flows (arrows)
(6) End with a final node.



* **State Diagram :-** A state diagram (also called state machine diagram) represents the states of an object & how it transitions from one state to another based on events.
It is useful for modeling reactive systems. (Systems that respond to external events)

Steps to draw a state diagram:-

(1) Identify the object whose states need to be modeled.
(2) Define all possible states of the object.
(3) Identify the events the trigger transitions between states.
(4) Mark the initial state (filled circle)
(5) Draw transitions (arrow) between states with event labels.
(6) Mark the final state (bullseye symbol)

Difference between Activity diagram & state diagram.

Aspect	Activity Diagram	State Diagram
Purpose	Represent workflow or sequence of activites	Represent states of an object & trasistion.
Focus	Focuses on actions & control flow	focuses on object states & events.
Start / End	Starts with an intial node & ends with a final node.	Starts with an initial states & ends with a final state.
Use case	Used for modeling business processes & System workflow	Used for modeling behaviour of Object in response to events
Nature	Similar to a flowchart	Similar to a finite state machine.

State diagram example -

States - New order -> Processing -> Shipped -> delivered
Transitions :-
Place order → New order.
Confirmation of Payment → Processing
Dispatch → Shipped
Customer Receives → Delivered.

<u>Result</u> :- The concepts of Activity Diagrams & state diagrams were studied. The diagrams were successfully drawn for the sample system, & the difference between them were understood.

Teacher's Signature